Applications: Growth and Decay

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Growth models**

A city's population is modeled by $P(t) = 12000(1.035)^t$, where t is in years.

a State the initial population and the annual growth rate. Initial population 12,000; growth rate 3.5% per year.

b Find the population after 10 years, to the nearest person. $P(10) = 12000(1.035)^{10} \approx 16{,}927$.

(a) Initial population 12,000; growth rate 3.5% per year.; (b) $P(10) = 12000(1.035)^{10} \approx 16{,}927$.

2 **[Foundational]** • **Target: Growth models**

An account earns 4% annual interest compounded continuously. How long does it take an investment to double? Give an exact expression and a decimal to two places.

$2 = e^{0.04t}$, so $t = \frac{\ln 2}{0.04} \approx 17.33$ years.

3 **[Foundational]** • **Target: Decay models**

Carbon-14 has a half-life of 5730 years: $A(t) = A_0\left(\frac{1}{2}\right)^{t/5730}$.

a What fraction of the original carbon-14 remains after 2000 years? Answer to three decimal places. $\left(\frac{1}{2}\right)^{2000/5730} \approx 0.785$.

b A sample retains 30% of its carbon-14. Estimate its age to the nearest hundred years. $0.3 = (0.5)^{t/5730}$ gives $t = 5730\,\frac{\ln 0.3}{\ln 0.5} \approx 10{,}000$ years.

(a) $\left(\frac{1}{2}\right)^{2000/5730} \approx 0.785$.; (b) $0.3 = (0.5)^{t/5730}$ gives $t = 5730\,\frac{\ln 0.3}{\ln 0.5} \approx 10{,}000$ years.

4 **[Foundational]** • **Target: Decay models**

A 200 g sample of a radioactive substance decays to 50 g in 12 hours, following $A(t) = 200e^{kt}$.

a Find the decay constant k (4 d.p.). $\frac{50}{200} = e^{12k}$, so $k = \frac{\ln(1/4)}{12} \approx -0.1155$.

b Find the half-life of the substance. $200 \to 50$ is two halvings in 12 h, so the half-life is exactly 6 hours (equivalently $\frac{\ln 2}{0.1155} \approx 6.0$ h).

(a) $\frac{50}{200} = e^{12k}$, so $k = \frac{\ln(1/4)}{12} \approx -0.1155$.; (b) $200 \to 50$ is two halvings in 12 h, so the half-life is exactly 6 hours (equivalently $\frac{\ln 2}{0.1155} \approx 6.0$ h).

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5 [Foundational] • Target: Decay models

Newton's law of cooling: coffee at $90°$C cools in a $20°$C room following $T(t) = 20 + 70e^{-0.1t}$ (t in minutes). Find the temperature after 10 minutes, to one decimal place.

$T(10) = 20 + 70e^{-1} \approx 20 + 25.75 = 45.8°$C.

6 [Foundational] • Target: Logistic growth (an introduction)

A lake is stocked with fish, and the population is modeled by $P(t) = \frac{5000}{1 + 9e^{-0.4t}}$, where t is in years.

a Find the initial population $P(0)$. $P(0) = \frac{5000}{1+9} = 500$.

b Find the carrying capacity (the value P approaches as $t \to \infty$). As $t \to \infty$, $e^{-0.4t} \to 0$, so $P \to \frac{5000}{1} = 5000$.

c Find $P(10)$, to the nearest fish. $P(10) = \frac{5000}{1 + 9e^{-4}} \approx \frac{5000}{1.1648} \approx 4292$.

(a) $P(0) = \frac{5000}{1+9} = 500$.; (b) As $t \to \infty$, $e^{-0.4t} \to 0$, so $P \to \frac{5000}{1} = 5000$.; (c) $P(10) = \frac{5000}{1 + 9e^{-4}} \approx \frac{5000}{1.1648} \approx 4292$.

7 [Foundational] • Target: Comparing models

Two towns start with 10,000 people. Town A grows at a constant rate of 300 people/year (linear); Town B grows at 3% per year (exponential). Find each town's population after 15 years, to the nearest person, and state which is larger.

Town A: $10000 + 300(15) = 14{,}500$. Town B: $10000(1.03)^{15} \approx 10000(1.5580) \approx 15{,}580$. Town B is larger.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Comparing models

A radioactive isotope decays from 80 g to 10 g over 3 half-lives. Find the half-life in terms of the decay time, and write the exponential decay model $A(t) = 80e^{kt}$, finding k if the total elapsed time is 15 years (4 d.p.).

$80 \to 10$ over 3 halvings ($80 \to 40 \to 20 \to 10$) means 3 half-lives $= 15$ years, so half-life $= 5$ years. Using $A(15) = 10$: $10 = 80e^{15k}$, so $k = \frac{\ln(1/8)}{15} \approx -0.1386$.

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9 [Intermediate] • Target: Visualizing the logistic curve

The fish population model $P(t) = \frac{5000}{1 + 9e^{-0.4t}}$ (from earlier) is graphed below. Use the graph to estimate where the population is growing fastest, then explain why this happens near the inflection point.

The curve is steepest near its midpoint, roughly where $P \approx 2500$ (half the carrying capacity), consistent with the logistic model's fastest growth occurring at $P = \frac{L}{2}$.

10 [Intermediate] • Target: Doubling time and half-life formulas

Derive a general formula for the doubling time T of a quantity growing as $A(t) = A_0 e^{kt}$, in terms of k.

Set $2A_0 = A_0 e^{kT}$: $2 = e^{kT} \Rightarrow \ln 2 = kT \Rightarrow T = \frac{\ln 2}{k}$.

11 [Intermediate] • Target: Doubling time and half-life formulas

Using the doubling-time formula derived above, find the doubling time for a population growing at a continuous rate of $k = 0.025$ per year, to one decimal place.

$T = \frac{\ln 2}{0.025} \approx \frac{0.6931}{0.025} \approx 27.7$ years.

12 [Intermediate] • Target: Half-life formula derivation

Derive a general formula for the half-life $T_{1/2}$ of a quantity decaying as $A(t) = A_0 e^{kt}$ (with $k < 0$), in terms of k.

Set $\frac{A_0}{2} = A_0 e^{kT_{1/2}}$: $\frac{1}{2} = e^{kT_{1/2}} \Rightarrow \ln\left(\frac{1}{2}\right) = kT_{1/2} \Rightarrow T_{1/2} = \frac{-\ln 2}{k}$ (positive since $k < 0$).

13 [Intermediate] • Target: Half-life formula derivation

A radioactive isotope has decay constant $k = -0.023$ per year. Use the formula above to find its half-life, to one decimal place.

$T_{1/2} = \frac{-\ln 2}{-0.023} = \frac{0.6931}{0.023} \approx 30.1$ years.

14 [Intermediate] • Target: Combining growth models with rates

A savings account earns continuous interest, growing from 5000 to 5512.50 riyals in 2 years. Find the annual interest rate r using $A = Pe^{rt}$, to two decimal places (as a percentage).

$5512.50 = 5000e^{2r} \Rightarrow 1.1025 = e^{2r} \Rightarrow 2r = \ln(1.1025) \approx 0.0976 \Rightarrow r \approx 0.0488$, i.e. approximately 4.88%.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Logistic growth (an introduction) • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Solve $\cos(2x) + \sin x = 0$ on $[0, 2\pi)$. (attempt 10)

Using $\cos(2x) = 1 - 2\sin^2 x$: $1 - 2\sin^2 x + \sin x = 0 \Rightarrow 2\sin^2 x - \sin x - 1 = 0$. Factoring: $(2\sin x + 1)(\sin x - 1) = 0$, so $\sin x = 1$ or $\sin x = -\frac{1}{2}$. This gives $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$.

16 [Challenging] • Target: Comparing models • Cross-Lesson: 3.03 Sum and Difference Formulas

Consider the exponential growth model $P(t) = P_0 e^{kt}$, where $P(t)$ is the population at time t and P_0 is the initial population. If the initial population is $P_0 = 500$ and after 10 years the population is $P(10) = 1200$, find the annual growth rate k. Round your answer to four decimal places.

Final Answer: 0.0434. We are given the exponential growth model $P(t) = P_0 e^{kt}$. We know that the initial population $P_0 = 500$ and that after 10 years, $P(10) = 1200$. We can substitute these values into the model: $1200 = 500e^{k \cdot 10}$ Divide both sides by 500: $\frac{1200}{500} = e^{10k}$ $2.4 = e^{10k}$ To solve for k, we take the natural logarithm of both sides: $\ln(2.4) = \ln(e^{10k})$ $\ln(2.4) = 10k$ Now, divide by 10: $k = \frac{\ln(2.4)}{10}$ Using a calculator, $\ln(2.4) \approx 0.8754687$..

17 [Challenging] • Target: Visualizing the logistic curve • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Consider the exponential growth model described by the function $P(t) = P_0 e^{kt}$, where $P(t)$ is the population at time t, P_0 is the initial population, and k is the growth constant. If the initial population P_0 is 500 and the population after 4 years is 2000, determine the value of the growth constant k to three decimal places.

Final Answer: 0.347. The population model is given by $P(t) = P_0 e^{kt}$. We are given that $P_0 = 500$. We are also given that $P(4) = 2000$. Substituting these values into the equation, we get: $2000 = 500e^{k \cdot 4}$ Divide both sides by 500: $4 = e^{4k}$ Take the natural logarithm of both sides: $\ln(4) = \ln(e^{4k})$ $\ln(4) = 4k$ Solve for k: $k = \frac{\ln(4)}{4}$ Using a calculator, $\ln(4) \approx 1.38629$. $k \approx \frac{1.38629}{4}$ $k \approx 0.34657$ Rounding to three decimal places.

18 [Challenging] • Target: Doubling time and half-life formulas • Cross-Lesson: 4.01 Circles and Parabolas

A population of bacteria, initially P_0 cells, grows according to the logistic model $P(t) = \frac{L}{1 + Ae^{-kt}}$, where t is in hours, L is the carrying capacity, and A and k are positive constants. At $t = 0$, the population is $P(0) = \frac{L}{3}$. After 2 hours, the population has doubled to $P(2) = \frac{2L}{3}$. Find the exact value of e^{2k} in terms of L and P_0.

Final Answer: $\frac{9}{4}$. The logistic model is given by $P(t) = \frac{L}{1 + Ae^{-kt}}$. At $t = 0$, $P(0) = \frac{L}{3}$. Substituting into the model: $\frac{L}{3} = \frac{L}{1 + Ae^{-k \cdot 0}}$ $\frac{L}{3} = \frac{L}{1 + A}$ $1 + A = 3$ $A = 2$. So the model becomes $P(t) = \frac{L}{1 + 2e^{-kt}}$. We are also given that $P(2) = \frac{2L}{3}$. Substituting into the model: $\frac{2L}{3} = \frac{L}{1 + 2e^{-2k}}$ $\frac{2}{3} = \frac{1}{1 + 2e^{-2k}}$ $2(1 + 2e^{-2k}) = 3$ $2 + 4e^{-2k} = 3$ $4e^{-2k} = 1$.

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19 [Challenging] • Target: Half-life formula derivation • Cross-Lesson: 4.02 Ellipses

The graph of the function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of the parameter a.

Final Answer: 2. We are given the function $f(x) = a \cdot b^x$ and two points it passes through: $(1, 12)$ and $(3, 108)$. Using the first point $(1, 12)$, we have: $12 = a \cdot b^1$ (Equation 1) Using the second point $(3, 108)$, we have: $108 = a \cdot b^3$ (Equation 2) To find a and b, we can divide Equation 2 by Equation 1: $\frac{108}{12} = \frac{a \cdot b^3}{a \cdot b^1}$ $9 = b^2$ Taking the square root of both sides, and assuming $b > 0$ for exponential functions: $b = 3$ Now, substitute the value of.

20 [Challenging] • Target: Combining growth models with rates • Cross-Lesson: 4.03 Hyperbolas

Consider the exponential function $f(x) = 2^x$. If $f(a) = 8$, what is the value of a? Show your complete algebraic work.

Final Answer: 3. The given function is $f(x) = 2^x$. We are given that $f(a) = 8$. Substituting a into the function, we get $2^a = 8$. To solve for a, we need to express 8 as a power of 2. Since $2^3 = 2 \times 2 \times 2 = 8$, we can rewrite the equation as $2^a = 2^3$. Since the bases are equal, the exponents must be equal. Therefore, $a = 3$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Population Growth Models

A bacterial culture doubles every 3 hours. If the initial count is 500, what is the population after 9 hours?

a. 4000 ✓

b. 1500

c. 2000

d. 8000

After 9 hours, $9/3 = 3$ doubling periods have elapsed. $P(9) = 500 \times 2^3 = 500 \times 8 = 4000$.

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22 [Foundational] • Target: Population Growth Models

A city has population $P(t) = 100000e^{0.02t}$, where t is in years. Find the continuous annual growth rate and doubling time.

a. **Rate** $= 2\%$**, Doubling time** ≈ 34.7 years ✓

b. Rate $= 20\%$, Doubling time ≈ 3.5 years

c. Rate $= 0.02\%$, Doubling time ≈ 50 years

d. Rate $= 2\%$, Doubling time ≈ 50 years

Growth constant $k = 0.02 = 2\%$. Doubling time $T = \frac{\ln 2}{k} = \frac{0.69315}{0.02} \approx 34.66 \approx 34.7$ years.

23 [Foundational] • Target: Population Growth Models

If a culture grows from 200 to 600 in 4 hours according to $P(t) = P_0 e^{kt}$, find k.

a. $k = \frac{\ln 3}{4} \approx 0.275$ ✓

b. $k = \frac{\ln 4}{3}$

c. $k = 3/4$

d. $k = \ln(3/4)$

$600 = 200e^{4k} \Rightarrow e^{4k} = 3 \Rightarrow 4k = \ln 3 \Rightarrow k = \frac{\ln 3}{4} \approx 0.275$.

24 [Foundational] • Target: Radioactive Decay and Half-Life

Carbon-14 has a half-life of 5730 years. What fraction of the original C-14 remains in a fossil after 17190 years?

a. 1/8 ✓

b. 1/4

c. 1/16

d. 1/3

Number of half-lives $= 17190/5730 = 3$. Remaining fraction $= (1/2)^3 = 1/8$.

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25 [Foundational] • Target: Radioactive Decay and Half-Life

State the formula relating decay constant $k > 0$ and half-life $t_{1/2}$ in $N(t) = N_0 e^{-kt}$.

a. $k = \frac{\ln 2}{t_{1/2}}$ ✓

b. $k = \frac{t_{1/2}}{\ln 2}$

c. $k = \frac{1}{2t_{1/2}}$

d. $k = \ln(2t_{1/2})$

$N_0/2 = N_0 e^{-kt_{1/2}} \Rightarrow e^{kt_{1/2}} = 2 \Rightarrow k = \frac{\ln 2}{t_{1/2}}$.

26 [Foundational] • Target: Radioactive Decay and Half-Life

A sample contains 80 mg of an isotope with a half-life of 10 days. How much remains after 25 days?

a. $80(0.5)^{2.5} \approx 14.14$ mg ✓

b. 20 mg

c. 10 mg

d. 15 mg

$N(25) = 80(1/2)^{25/10} = 80(1/2)^{2.5} = \frac{80}{4\sqrt{2}} = \frac{20}{\sqrt{2}} = 10\sqrt{2} \approx 14.14$ mg.

27 [Foundational] • Target: Newton's Law of Cooling

State Newton's Law of Cooling formula for temperature $T(t)$ with ambient medium T_m.

a. $T(t) = T_m + (T_0 - T_m)e^{-kt}$ ✓

b. $T(t) = T_0 e^{-kt} + T_m$

c. $T(t) = T_m(1 - e^{-kt})$

d. $T(t) = (T_0 - T_m)e^{kt}$

The rate of cooling is proportional to the temperature difference:
$\frac{dT}{dt} = -k(T - T_m) \Rightarrow T(t) = T_m + (T_0 - T_m)e^{-kt}$.

28 [Foundational] • Target: Newton's Law of Cooling

A hot cup of coffee at $190\,^\circ$F is placed in a $70\,^\circ$F room. What is the initial temperature difference $T_0 - T_m$?

a. $120\,^\circ$F ✓

b. $190\,^\circ$F

c. $70\,^\circ$F

d. $260\,^\circ$F

$T_0 - T_m = 190 - 70 = 120\,^\circ$F.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Newton's Law of Cooling**

If the coffee cools from $190\,^\circ\text{F}$ to $130\,^\circ\text{F}$ in 10 minutes in the $70\,^\circ\text{F}$ room, find k.

a. $k = \frac{\ln 2}{10} \approx 0.0693\ \text{min}^{-1}$ ✓

b. $k = 0.1$

c. $k = \frac{\ln 3}{10}$

d. $k = 0.05$

$130 = 70 + 120e^{-10k} \Rightarrow 60 = 120e^{-10k} \Rightarrow e^{-10k} = 1/2 \Rightarrow 10k = \ln 2 \Rightarrow k = \frac{\ln 2}{10}$.

30 **[Intermediate]** • **Target: Logarithmic Scales (pH, Richter, Decibels)**

The pH of a solution is defined as $\text{pH} = -\log_{10}[\text{H}^+]$. What is the pH if $[\text{H}^+] = 10^{-4}$ M?

a. 4 ✓

b. -4

c. 10

d. 14

$\text{pH} = -\log_{10}(10^{-4}) = -(-4) = 4$.

31 **[Intermediate]** • **Target: Logarithmic Scales (pH, Richter, Decibels)**

The Richter magnitude scale is $M = \log_{10}\left(\frac{I}{S}\right)$. How many times more intense is an earthquake of magnitude 7 than one of magnitude 4?

a. 1000 **times (**10^3**)** ✓

b. 3 times

c. 30 times

d. 10000 times

Difference in magnitude $\Delta M = 7 - 4 = 3$. Ratio of intensities $= 10^3 = 1000$.

32 **[Intermediate]** • **Target: Logarithmic Scales (pH, Richter, Decibels)**

Sound intensity level in decibels is $\beta = 10\log_{10}\left(\frac{I}{I_0}\right)$. If the intensity I is multiplied by 100, how many decibels are added?

a. +20 dB ✓

b. +100 dB

c. +10 dB

d. +2 dB

$\Delta\beta = 10\log_{10}(100) = 10(2) = 20$ dB.

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33 [Intermediate] • Target: Growth models

Consider the exponential function $f(x) = 2^x$. What is the value of $f(3)$?

a. 6

b. **8 ✓**

c. 9

d. 5

The function is given by $f(x) = 2^x$. To find $f(3)$, we substitute $x = 3$ into the function: $f(3) = 2^3$. Calculating the power, $2^3 = 2 \times 2 \times 2 = 8$. Therefore, the value of $f(3)$ is 8.

34 [Intermediate] • Target: Decay models

Consider the exponential function $f(x) = 3^x$. What is the value of $f(2)$?

a. 6

b. **9 ✓**

c. 8

d. 5

The function is defined as $f(x) = 3^x$. To find the value of $f(2)$, we substitute $x = 2$ into the function: $f(2) = 3^2$. Calculating 3^2 gives $3 \times 3 = 9$. Therefore, $f(2) = 9$.

35 [Intermediate] • Target: Logistic growth (an introduction)

A population of 100 grows at 2% per year, modeled by $P(t) = 100(1 + 0.02)^t$. Approximately how many years until the population reaches 1.5× its initial size (i.e. $P = 1.5 \cdot 100$)?

a. 22.5 years

b. **20.5 years ✓**

c. 26.7 years

d. 18.5 years

Solve $1.5 = (1 + 0.02)^t$: $t = \frac{\ln 1.5}{\ln(1 + 0.02)} \approx 20.5$ years.

36 [Intermediate] • Target: Comparing models

A population of 500 grows at 3% per year, modeled by $P(t) = 500(1 + 0.03)^t$. Approximately how many years until the population reaches 1.5× its initial size (i.e. $P = 1.5 \cdot 500$)?

a. 15.7 years

b. 17.8 years

c. 11.7 years

d. **13.7 years ✓**

Solve $1.5 = (1 + 0.03)^t$: $t = \frac{\ln 1.5}{\ln(1 + 0.03)} \approx 13.7$ years.

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37 [Intermediate] • Target: Visualizing the logistic curve

A population of 500 grows at 2% per year, modeled by $P(t) = 500(1 + 0.02)^t$. Approximately how many years until the population reaches 1.5× its initial size (i.e. $P = 1.5 \cdot 500$)?

a. 22.5 years

b. **20.5 years ✓**

c. 26.7 years

d. 18.5 years

Solve $1.5 = (1 + 0.02)^t$: $t = \frac{\ln 1.5}{\ln(1 + 0.02)} \approx 20.5$ years.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Doubling time and half-life formulas • Cross-Lesson: 4.01 Circles and Parabolas

A population of 1000 grows at 2% per year, modeled by $P(t) = 1000(1 + 0.02)^t$. Approximately how many years until the population reaches 1.5× its initial size (i.e. $P = 1.5 \cdot 1000$)?

a. 26.7 years

b. 18.5 years

c. 22.5 years

d. **20.5 years ✓**

Solve $1.5 = (1 + 0.02)^t$: $t = \frac{\ln 1.5}{\ln(1 + 0.02)} \approx 20.5$ years.

39 [Challenging] • Target: Half-life formula derivation • Cross-Lesson: 4.02 Ellipses

A population of 200 grows at 2% per year, modeled by $P(t) = 200(1 + 0.02)^t$. Approximately how many years until the population reaches 1.5× its initial size (i.e. $P = 1.5 \cdot 200$)?

a. 22.5 years

b. 18.5 years

c. 26.7 years

d. **20.5 years ✓**

Solve $1.5 = (1 + 0.02)^t$: $t = \frac{\ln 1.5}{\ln(1 + 0.02)} \approx 20.5$ years.

40 [Challenging] • Target: Combining growth models with rates • Cross-Lesson: 4.03 Hyperbolas

A population of 1000 grows at 3% per year, modeled by $P(t) = 1000(1 + 0.03)^t$. Approximately how many years until the population reaches 1.5× its initial size (i.e. $P = 1.5 \cdot 1000$)?

a. 17.8 years

b. 15.7 years

c. **13.7 years ✓**

d. 11.7 years

Solve $1.5 = (1 + 0.03)^t$: $t = \frac{\ln 1.5}{\ln(1 + 0.03)} \approx 13.7$ years.

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41 [Challenging] • Target: Population Growth Models • Cross-Lesson: 5.01 Vectors: Components and Operations

A population of 500 grows at 5% per year, modeled by $P(t) = 500(1 + 0.05)^t$. Approximately how many years until the population reaches 1.5× its initial size (i.e. $P = 1.5 \cdot 500$)?

a. 8.3 years ✓

b. 10.8 years

c. 6.3 years

d. 10.3 years

Solve $1.5 = (1 + 0.05)^t$: $t = \frac{\ln 1.5}{\ln(1 + 0.05)} \approx 8.3$ years.

42 [Challenging] • Target: Radioactive Decay and Half-Life • Cross-Lesson: 5.02 The Dot Product

A population of 500 grows at 8% per year, modeled by $P(t) = 500(1 + 0.08)^t$. Approximately how many years until the population reaches 1.5× its initial size (i.e. $P = 1.5 \cdot 500$)?

a. 5.3 years ✓

b. 7.3 years

c. 3.3 years

d. 6.9 years

Solve $1.5 = (1 + 0.08)^t$: $t = \frac{\ln 1.5}{\ln(1 + 0.08)} \approx 5.3$ years.

43 [Challenging] • Target: Newton's Law of Cooling • Cross-Lesson: 5.03 Parametric Equations

A population of 200 grows at 8% per year, modeled by $P(t) = 200(1 + 0.08)^t$. Approximately how many years until the population reaches 1.5× its initial size (i.e. $P = 1.5 \cdot 200$)?

a. 5.3 years ✓

b. 6.9 years

c. 7.3 years

d. 3.3 years

Solve $1.5 = (1 + 0.08)^t$: $t = \frac{\ln 1.5}{\ln(1 + 0.08)} \approx 5.3$ years.

44 [Challenging] • Target: Logarithmic Scales (pH, Richter, Decibels) • Cross-Lesson: 6.01 Polar Coordinates and Conversions

A population of 200 grows at 5% per year, modeled by $P(t) = 200(1 + 0.05)^t$. Approximately how many years until the population reaches 1.5× its initial size (i.e. $P = 1.5 \cdot 200$)?

a. 10.3 years

b. 6.3 years

c. 10.8 years

d. 8.3 years ✓

Solve $1.5 = (1 + 0.05)^t$: $t = \frac{\ln 1.5}{\ln(1 + 0.05)} \approx 8.3$ years.

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45 [Challenging] • Target: Growth models • Cross-Lesson: 6.02 Polar Equations and Graphs

A population of 100 grows at 5% per year, modeled by $P(t) = 100(1 + 0.05)^t$. Approximately how many years until the population reaches 1.5× its initial size (i.e. $P = 1.5 \cdot 100$)?

a. 6.3 years

b. 10.3 years

c. **8.3 years ✓**

d. 10.8 years

Solve $1.5 = (1 + 0.05)^t$: $t = \frac{\ln 1.5}{\ln(1 + 0.05)} \approx 8.3$ years.

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